

HOMEWORK 7
INTRODUCTION TO COMPLEX ANALYSIS
(MATH 4230-40), SPRING 2022

SECTION 2.4 - CAUCHY'S INTEGRAL FORMULA

1. Evaluate the following integrals:

(a) $\int_{\gamma} \frac{z^2 - 1}{z^2 + 1} dz$, where γ is a circle of radius 2, centered at 0.

(b) $\int_{\gamma} \frac{\sin e^z}{z} dz$, where γ is the unit circle.

2. Let f be entire and assume that $|f(z)| \leq M|z|^n$ for large $|z|$, for a constant M , and some non-negative integer n . Show that f is a polynomial of degree less than or equal to n .

3. Consider the function $f(z) = \frac{1}{z^2}$.

(a) Notice that $\int_{\gamma} f(z) dz = 0$ for all closed contours γ not passing through the origin, but is not analytic at $z = 0$. Does this statement contradict Morera's Theorem?

(b) Notice that f is bounded as $|z| \rightarrow \infty$, but is not a constant. Does this statement contradict Liouville's theorem?